Grade 4 science syllabus

PART 1: GRADE 4 SCIENCE – UNITS & TOPICS

UNIT 1: My Body

Topics

Body Parts

Internal Organs

Functions of Organs

Hygiene

Healthy Habits

UNIT 2: Matter & Materials

Topics

Solids

Liquids

Changes by Heat and Cold

Types of Materials

Properties (hard/soft)

UNIT 3: Food

Topics

Healthy Food

Junk Food

Food Groups

Cooking Methods

Balanced Diet

UNIT 4: Plants

Topics

Parts of Plant

Needs of Plants

Plant Growth

Uses of Plants

Types of Plants

UNIT 5: Solar Family

Topics

Sun

Planets

Planet Order

Earth

Day & Night

UNIT 1: MY BODY – FULL GAME PROMPTS

GAME 1: Body Parts Tap

Game Title: Body Parts Tap

Topic: Body Parts

Objective:

Help kids identify external body parts.

LOGIN SCREEN:

- Enter Name

- Avatar selection (cartoon boy/girl)

- Start button

- Background: colorful body theme

GAMEPLAY LOGIC:

- Show cartoon human body

- Highlight question: “Tap the HEART”

- Student taps correct part

LEVELS:

Level 1:

- Basic parts (eye, nose, hand)

Level 2:

- More parts (heart, lungs)

Level 3:

- Timer challenge

UI:

- Cartoon human body

- Glow effect on correct part

- Big tappable areas

SOUND:

- Correct → “Great job!” + sparkle sound

- Wrong → “Try again!” soft sound

SCORING:

- +10 per correct

- Streak increases

FINAL DASHBOARD:

- Total Score

- Accuracy %

- Best Streak

- Stars

- Replay button

GAME 2: Inside My BodyGame Title: Inside My Body

Topic: Internal Organs

Objective:

Help kids recognize internal organs.

GAMEPLAY LOGIC:

- Show body outline (X-ray style)

- Ask: “Find the lungs”

- Tap correct organ

LEVELS:

L1: Heart, lungs

L2: Add brain, stomach

L3: Timer challenge

UI:

- X-ray animation

- Organs glowing

SOUND:

- Correct → “Awesome!”

- Wrong → soft retry

SCORING + DASHBOARD:

(Same system)

GAME 3: Organ Function Match

Game Title: Organ Function Match

Topic: Organ Functions

Objective:

Match organs with their functions.

GAMEPLAY LOGIC:

- Show: Heart, Lungs

- Show: Pumps blood, Helps breathing

- Drag to match

LEVELS:

L1: 2 pairs

L2: 3–4 pairs

L3: Timer challenge

UI:

- Card matching style

- Drag-and-drop

SOUND:

- Correct → “Perfect match!”

- Wrong → retry

SCORING + DASHBOARD:

(Same system)

GAME 4: Hygiene Hero

Game Title: Hygiene Hero

Topic: Hygiene

Objective:

Teach good hygiene habits.

GAMEPLAY LOGIC:

- Show actions (brushing, dirty hands)

- Ask: “Choose clean habit”

- Tap correct option

LEVELS:

L1: Clear habits

L2: Mixed actions

L3: Timer challenge

UI:

- Cartoon kids

- Before/after scenes

SOUND:

- Correct → “Clean and healthy!”

- Wrong → soft retry

SCORING + DASHBOARD:

(Same system)

GAME 5: Healthy Habits Builder

Game Title: Healthy Habits Builder

Topic: Healthy Habits

Objective:

Help kids choose daily healthy habits.

GAMEPLAY LOGIC:

- Show actions (eat, sleep, play, brush)
- Ask: “Pick healthy habits”
- Multiple selection

LEVELS:

L1: Simple habits

L2: Mixed options

L3: Timer challenge

UI:

- Daily routine timeline

- Animated icons

SOUND:

- Correct → “Great routine!”

- Wrong → retry

SCORING + DASHBOARD:

(Same system)

Game Title: Healthy Habits Builder

Topic: Healthy Habits

Objective:

Help kids choose daily healthy habits.

GAMEPLAY LOGIC:

- Show actions (eat, sleep, play, brush)

- Ask: “Pick healthy habits”

- Multiple selection

LEVELS:

L1: Simple habits

L2: Mixed options

L3: Timer challenge

UI:

- Daily routine timeline

- Animated icons

SOUND:

- Correct → “Great routine!”

- Wrong → retry

SCORING + DASHBOARD:

(Same system)

Game Title: Healthy Habits Builder

Topic: Healthy Habits

Objective:

Help kids choose daily healthy habits.

GAMEPLAY LOGIC:

- Show actions (eat, sleep, play, brush)
- Ask: “Pick healthy habits”
- Multiple selection

LEVELS:

L1: Simple habits

L2: Mixed options

L3: Timer challenge

UI:

- Daily routine timeline
- Animated icons

SOUND:

- Correct → “Great routine!”

- Wrong → retry

SCORING + DASHBOARD:

(Same system)

UNIT 2: MATTER & MATERIALS – FULL GAME PROMPTS

GAME 6: Solid or Liquid Sort

Game Title: Solid or Liquid Sort

Topic: Solids & Liquids

Objective:

Help students identify solids and liquids.

LOGIN SCREEN:

- Name input

- Avatar selection

- Start button

- Background: colorful science lab

GAMEPLAY LOGIC:

- Show objects (stone, milk, juice, book)
- Two boxes: SOLID / LIQUID
- Student drags objects into correct box

LEVELS:

Level 1:

- Clear examples

Level 2:

- Mixed objects

Level 3:

- Timer challenge

UI:

- Lab theme
- Drag-and-drop boxes
- Liquid animation

SOUND:

- Drop sound
- Correct → “Great sorting!”
- Wrong → soft retry

SCORING:

- +10 per correct
- Streak

FINAL DASHBOARD:

- Total Score
- Accuracy %
- Best Streak
- Stars
- Replay button

GAME 7: Liquid Finder

Game Title: Liquid Finder

Topic: Liquids

Objective:

Help students identify liquid items.

GAMEPLAY LOGIC:

- Show multiple objects (water, oil, stone, chair)
- Ask: “Tap all liquids”
- Student selects correct ones

LEVELS:

L1: Clear liquids

L2: Mixed items

L3: Timer challenge

UI:

- Water-themed background
- Highlight animation

SOUND:

- Water splash sound

- Correct → “Nice!”

SCORING + DASHBOARD:

(Same system)

GAME 8: Hot or Cold Change

Game Title: Hot or Cold Change

Topic: Changes in Matter

Objective:

Teach how materials change with heat and cold.

GAMEPLAY LOGIC:

- Show example: Ice → ?

- Options: Water / Stone / Wood

- Student selects correct change

LEVELS:

L1: Ice → water

L2: Water → steam

L3: Mixed + timer

UI:

- Fire and ice effects

- Transformation animation

SOUND:

- Melting sound

- Correct → “Amazing!”

SCORING + DASHBOARD:

(Same system)

GAME 9: Material Match

Game Title: Material Match

Topic: Types of Materials

Objective:

Help students match objects with materials.

GAMEPLAY LOGIC:

- Show object (chair, bottle, spoon)

- Show materials (wood, plastic, metal)

- Drag to match

LEVELS:

L1: 2 options

L2: 3 options

L3: Timer challenge

UI:

- Real-world object cards

- Drag-and-drop

SOUND:

- Correct → “Perfect match!”

SCORING + DASHBOARD:

(Same system)

GAME 10: Hard or Soft Touch

Game Title: Hard or Soft Touch

Topic: Properties of Materials

Objective:

Help students identify material properties.

GAMEPLAY LOGIC:

- Show object (rock, sponge, pillow)
- Ask: “Hard or Soft?”
- Student taps correct option

LEVELS:

L1: Clear examples

L2: Mixed objects

L3: Timer challenge

UI:

- Touch-based icons

- Smooth animations

SOUND:

- Tap sound

- Correct → “Great!”

SCORING + DASHBOARD:

(Same system)

UNIT 3: FOOD – FULL GAME PROMPTS

GAME 11: Healthy Plate Builder

Game Title: Healthy Plate Builder

Topic: Healthy Food

Objective:

Help students choose healthy foods to build a balanced plate.

LOGIN SCREEN:

- Name input

- Avatar selection

- Start button

- Background: colorful kitchen

GAMEPLAY LOGIC:

- Show empty plate

- Show food items (fruits, vegetables, junk food)

- Student drags healthy items to plate

LEVELS:

Level 1:

- Clear healthy foods

Level 2:

- Mixed foods

Level 3:

- Timer challenge

UI:

- Plate layout

- Food animations

SOUND:

- Correct → “Yummy and healthy!”

- Wrong → soft retry

SCORING:

- +10 per correct

- Streak

FINAL DASHBOARD:

- Total Score

- Accuracy %

- Best Streak

- Stars

- Replay button

GAME 12: Junk vs Healthy

Game Title: Junk vs Healthy

Topic: Food Types

Objective:

Help students differentiate healthy and junk food.

GAMEPLAY LOGIC:

- Show food item (pizza, apple, chips)
- Options: HEALTHY / JUNK
- Student taps correct choice

LEVELS:

L1: Clear items

L2: Mixed foods

L3: Timer challenge

UI:

- Bright food cards
- Tap animation

SOUND:

- Correct → “Great choice!”

- Wrong → retry sound

SCORING + DASHBOARD:

(Same system)

GAME 13: Food Group SortGame Title: Food Group Sort

Topic: Food Groups

Objective:

Help students group foods into categories.

GAMEPLAY LOGIC:

- Show foods (rice, egg, fruits)
- Boxes: Energy / Body-building / Protective
- Drag to correct group

LEVELS:

L1: 2 groups

L2: 3 groups

L3: Timer challenge

UI:

- Color-coded boxes

- Drag-and-drop

SOUND:

- Drop sound

- Correct → “Perfect!”

SCORING + DASHBOARD:

(Same system)

GAME 14: Cooking Method Match

Game Title: Cooking Method Match

Topic: Cooking Methods

Objective:

Help students match food with cooking methods.

GAMEPLAY LOGIC:

- Show food (rice, bread, egg)

- Show methods (boiling, baking, frying)

- Drag to match

LEVELS:

L1: Easy matches

L2: More options

L3: Timer challenge

UI:

- Kitchen theme

- Animated cooking icons

SOUND:

- Cooking sound

- Correct → “Well done!”

SCORING + DASHBOARD:

(Same system)

GAME 15: Balanced Meal Builder

Game Title: Balanced Meal Builder

Topic: Balanced Diet

Objective:

Help students create a balanced meal.

GAMEPLAY LOGIC:

- Show plate sections
- Categories: fruits, vegetables, grains, protein
- Student fills all correctly

LEVELS:

L1: Simple plate

L2: More categories

L3: Timer challenge

UI:

- Sectioned plate

- Smooth animations

SOUND:

- Correct → “Perfect meal!”

SCORING + DASHBOARD:

(Same system)

UNIT 4: PLANTS – FULL GAME PROMPTS

GAME 16: Plant Parts Puzzle

Game Title: Plant Parts Puzzle

Topic: Parts of Plant

Objective:

Help students identify parts of a plant.

LOGIN SCREEN:

- Name input

- Avatar selection

- Start button

- Background: garden theme

GAMEPLAY LOGIC:

- Show plant image

- Show labels: root, stem, leaf, flower

- Student drags labels to correct parts

LEVELS:

Level 1:

- 3 parts (root, stem, leaf)

Level 2:

- Add flower

Level 3:

- Timer challenge

UI:

- Bright garden scene

- Drag-and-drop labels

- Highlight correct placement

SOUND:

- Drop sound

- Correct → “Great job!”

SCORING:

- +10 per correct

- Streak

FINAL DASHBOARD:

- Total Score

- Accuracy %

- Best Streak

- Stars

- Replay button

GAME 17: What Plants Need

Game Title: What Plants Need

Topic: Needs of Plants

Objective:

Help students understand what plants need to grow.

GAMEPLAY LOGIC:

- Show plant
- Options: water, sunlight, air, toys
- Student selects correct needs

LEVELS:

Level 1:

- Basic needs

Level 2:

- Mixed options

Level 3:

- Timer challenge

UI:

- Garden background

- Animated sun and water

SOUND:

- Water sound

- Correct → “Perfect!”

SCORING + DASHBOARD:

(Same system)

GAME 18: Grow the Plant

Game Title: Grow the Plant

Topic: Plant Growth

Objective:

Show how plants grow step by step.

GAMEPLAY LOGIC:

- Start with seed

- Provide items (water, sunlight)

- Plant grows step by step

LEVELS:

Level 1:

- Simple growth

Level 2:

- Multiple steps

Level 3:

- Timer challenge

UI:

- Growth animation

- Smooth transitions

SOUND:

- Magical grow sound

- Correct → “Amazing!”

SCORING + DASHBOARD:

(Same system)

GAME 19: Plant Uses Match

Game Title: Plant Uses Match

Topic: Uses of Plants

Objective:

Help students understand uses of plants.

GAMEPLAY LOGIC:

- Show plant/object (tree, cotton plant)
- Show uses (wood, clothes, food)
- Student matches correctly

LEVELS:

Level 1:

- Easy matches

Level 2:

- More options

Level 3:

- Timer challenge

UI:

- Nature cards

- Drag-and-drop

SOUND:

- Correct → “Well done!”

SCORING + DASHBOARD:

(Same system)

GAME 20: Plant Type Finder

Game Title: Plant Type Finder

Topic: Types of Plants

Objective:

Help students identify types of plants.

GAMEPLAY LOGIC:

- Show plant image
- Options: Tree / Shrub / Herb
- Student selects correct type

LEVELS:

Level 1:

- Clear examples

Level 2:

- Mixed plants

Level 3:

- Timer challenge

UI:

- Forest scene

- Bright visuals

SOUND:

- Nature sound

- Correct → “Great!”

SCORING + DASHBOARD:

(Same system)

UNIT 5: SOLAR FAMILY – FULL GAME PROMPTS

GAME 21: Sun or Not

Game Title: Sun or Not

Topic: Sun

Objective:

Help students identify the Sun among space objects.

LOGIN SCREEN:

- Name input

- Avatar selection (space theme)

- Start button

- Background: galaxy

GAMEPLAY LOGIC:

- Show objects (Sun, Moon, planets, stars)

- Ask: “Tap the Sun”

- Student selects correct

LEVELS:

Level 1:

- Clear images

Level 2:

- Mixed objects

Level 3:

- Timer challenge

UI:

- Space theme

- Glowing Sun animation

SOUND:

- Space sound

- Correct → “Bright answer!”

- Wrong → soft retry

SCORING:

- +10 per correct

- Streak

FINAL DASHBOARD:

- Total Score

- Accuracy %

- Best Streak

- Stars

- Replay button

GAME 22: Planet Match

Game Title: Planet Match

Topic: Planets

Objective:

Help students identify planets by name and appearance.

GAMEPLAY LOGIC:

- Show planet image
- Options: Earth, Mars, Jupiter
- Student selects correct name

LEVELS:

Level 1:

- Easy planets

Level 2:

- More planets

Level 3:

- Timer challenge

UI:

- Planet cards

- Rotating animations

SOUND:

- Space click sound

- Correct → “Great!”

SCORING + DASHBOARD:

(Same system)

GAME 23: Planet Order Game

Game Title: Planet Order Game

Topic: Order of Planets

Objective:

Teach students the order of planets from the Sun.

GAMEPLAY LOGIC:

- Show planets randomly

- Student arranges from Sun outward

LEVELS:

Level 1:

- 4 planets

Level 2:

- 6 planets

Level 3:

- All planets + timer

UI:

- Orbit lines animation

- Drag-and-drop planets

SOUND:

- Orbit sound

- Correct → “Perfect order!”

SCORING + DASHBOARD:

(Same system)

Game Title: Planet Order Game

Topic: Order of Planets

Objective:

Teach students the order of planets from the Sun.

GAMEPLAY LOGIC:

- Show planets randomly

- Student arranges from Sun outward

LEVELS:

Level 1:

- 4 planets

Level 2:

- 6 planets

Level 3:

- All planets + timer

UI:

- Orbit lines animation

- Drag-and-drop planets

SOUND:

- Orbit sound

- Correct → “Perfect order!”

SCORING + DASHBOARD:

(Same system)

GAME 24: Earth Explorer

Game Title: Earth Explorer

Topic: Earth

Objective:

Help students learn about Earth features.

GAMEPLAY LOGIC:

- Show Earth
- Ask: “Select water / land / air”
- Student taps correct features

LEVELS:

Level 1:

- Basic features

Level 2:

- Mixed options

Level 3:

- Timer challenge

UI:

- Rotating globe
- Highlight selection

SOUND:

- Nature sound

- Correct → “Awesome!”

SCORING + DASHBOARD:

(Same system)

GAME 25: Day or Night

Game Title: Day or Night

Topic: Day & Night

Objective:

Help students understand day and night concept.

GAMEPLAY LOGIC:

- Show scene (sun / moon / sky)

- Ask: “Is it day or night?”

- Student selects correct answer

LEVELS:

Level 1:

- Clear scenes

Level 2:

- Mixed visuals

Level 3:

- Timer challenge

UI:

- Day/night switch animation

- Sky transitions

SOUND:

- Day birds / night crickets

- Correct → “Correct!”

SCORING + DASHBOARD:

(Same system)